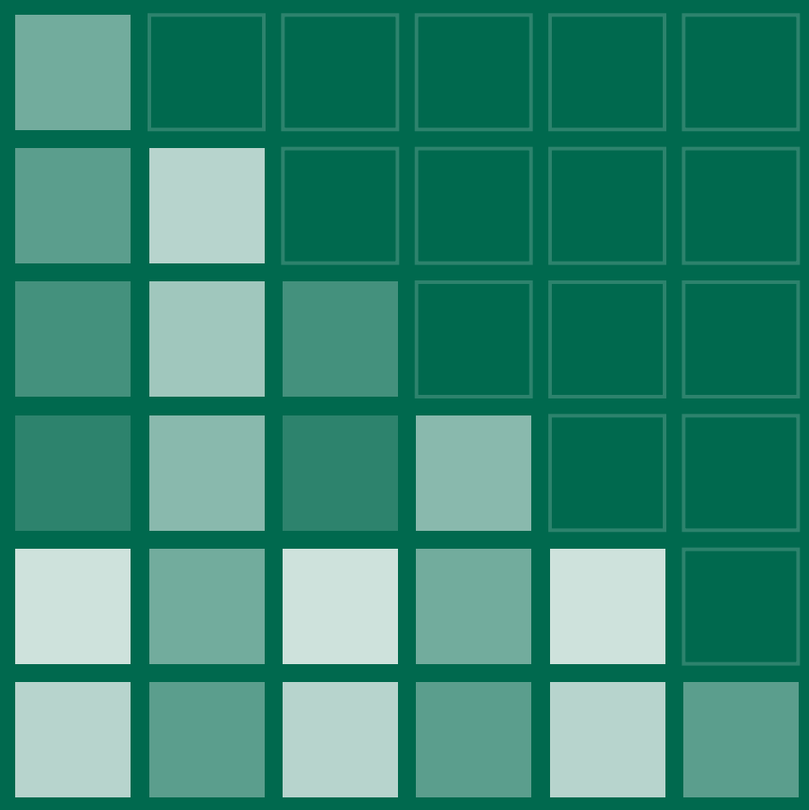
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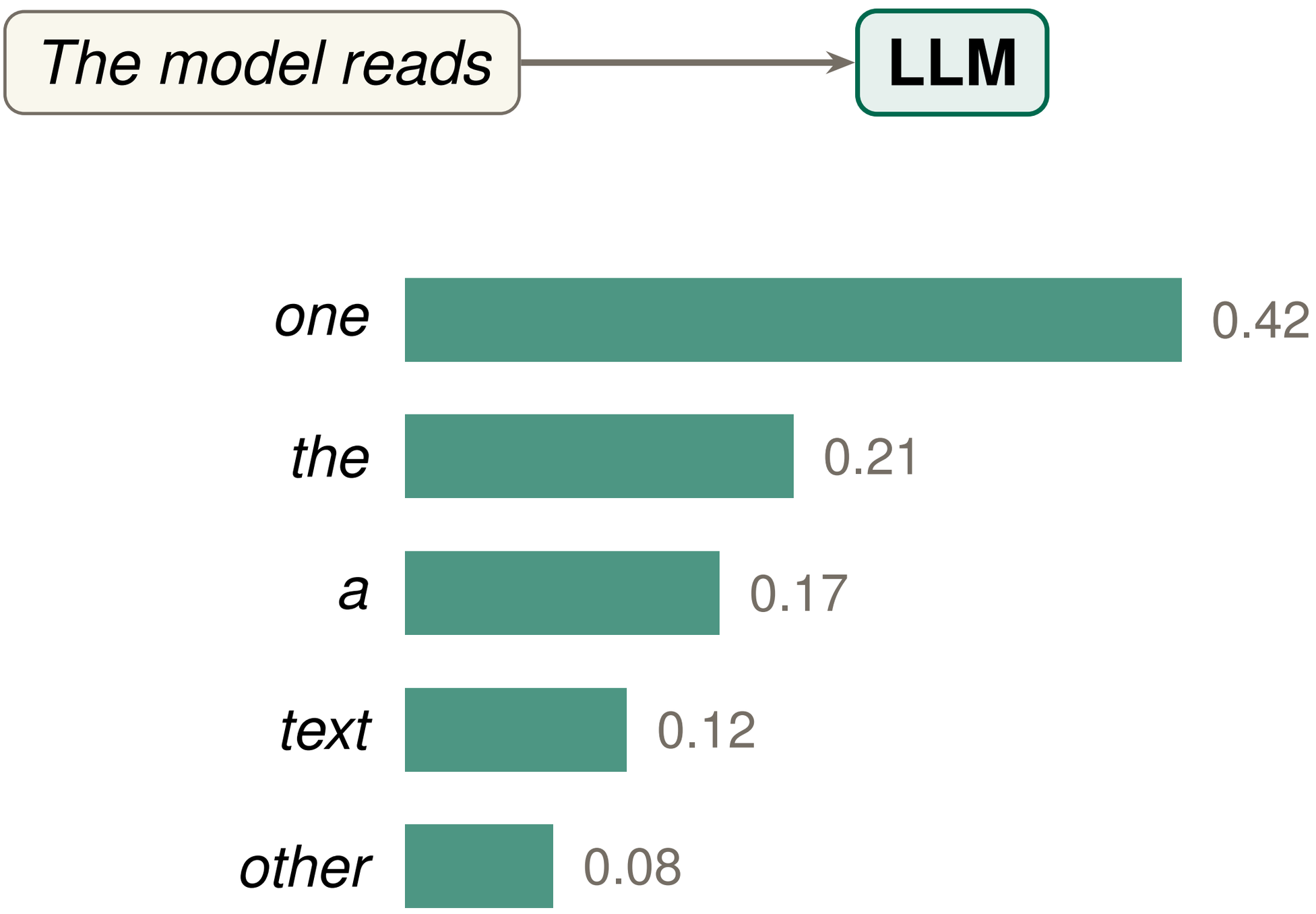
Every word an AI writes is one math function.

The decoder-only Transformer,
explained in 7 slides.

The *model* *reads* **?** $\rightarrow$ p(next word)

It’s a next-word guesser.

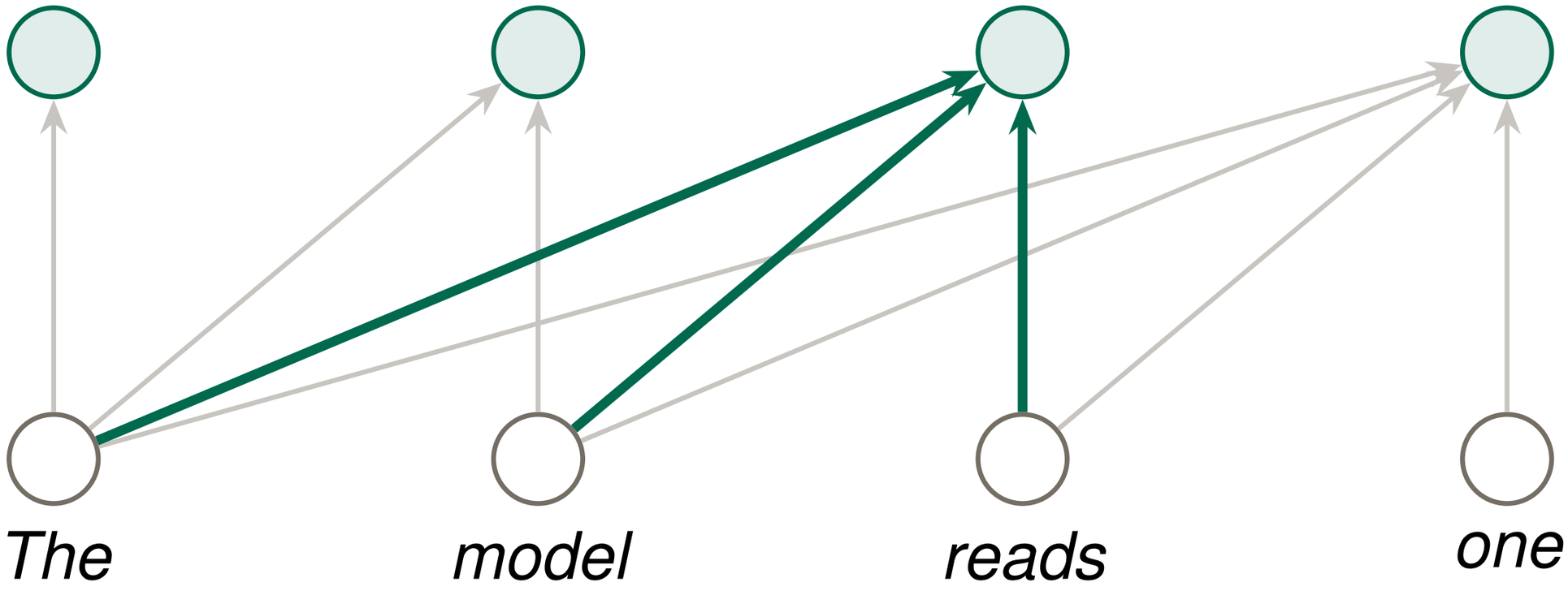
Input: the words so far.
Output: a probability for **every** possible next word.



Write: pick a word, add it, repeat.

Each word looks back. Never forward.

“reads” looks at 3 words



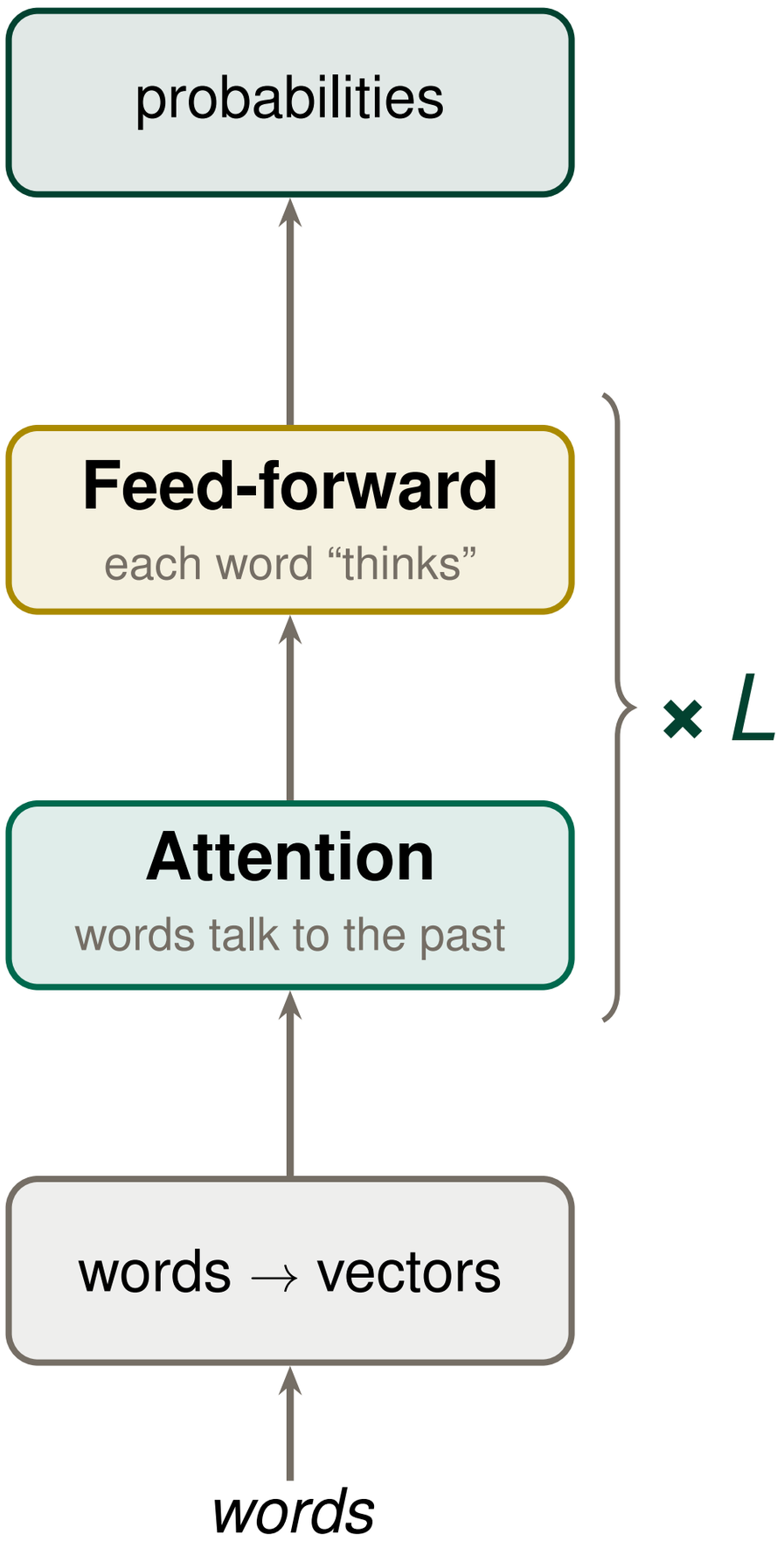
0	$-\infty$	$-\infty$	$-\infty$
0	0	$-\infty$	$-\infty$
0	0	0	$-\infty$
0	0	0	0

The **mask** puts $-\infty$ on the future.

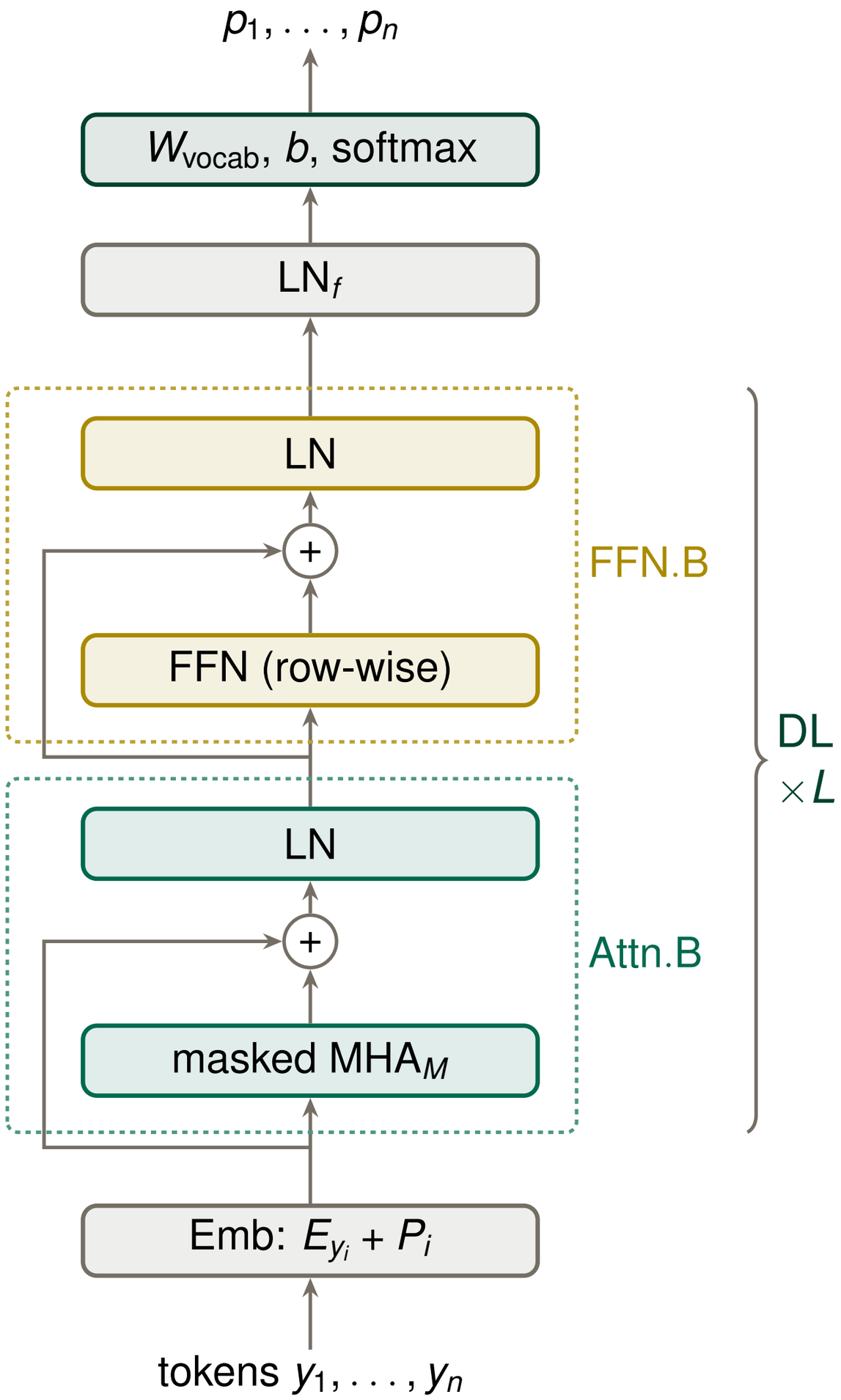
So the model can’t cheat by reading the answer.

One block. Repeated.

IN WORDS



IN MAPS



+ “skips” (the + arrows) let learning signals reach the bottom.

Attention = a smart weighted average.

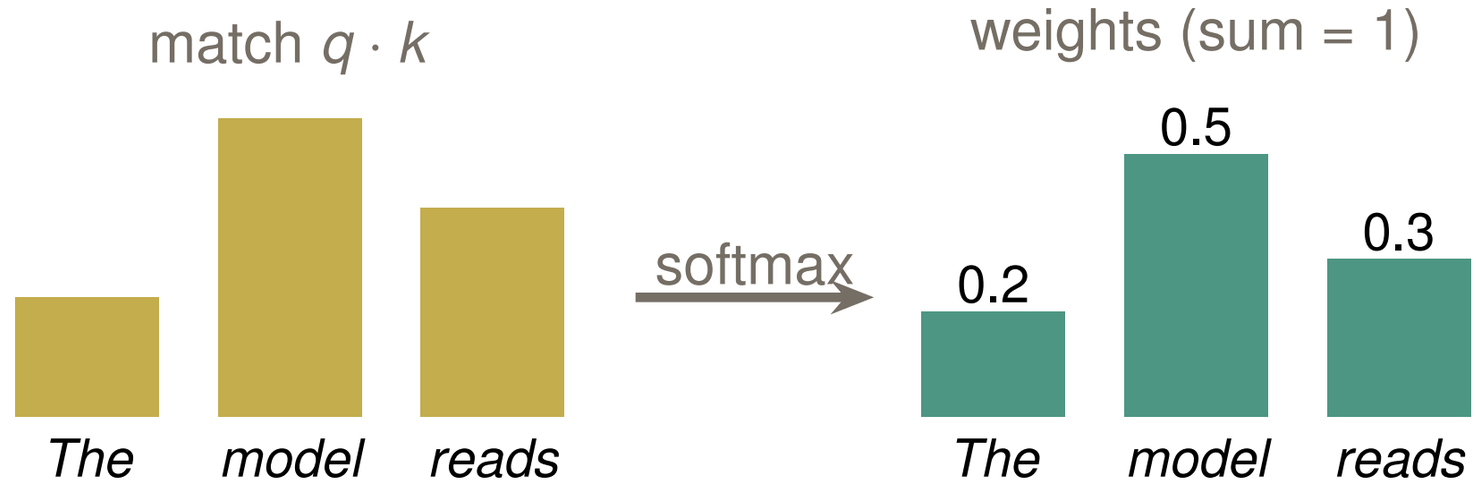
Query (Q)
what am I looking for?

Key (K)
what do I offer?

Value (V)
what I hand over

For the word *reads*:

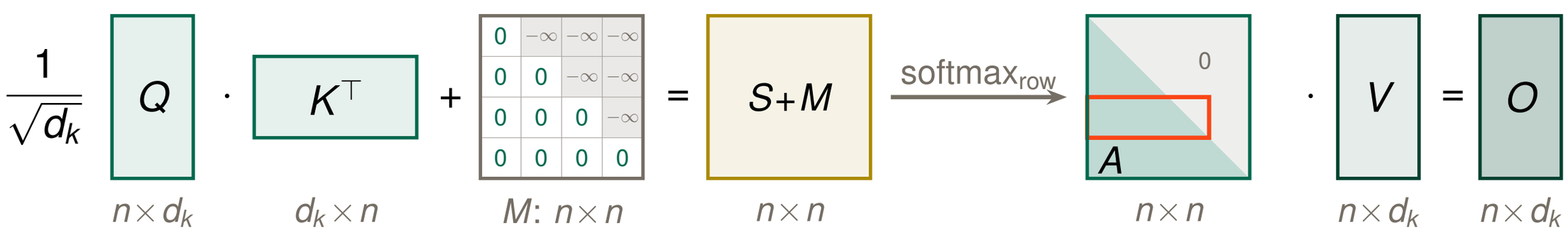
Each word pulls in information from the words that matter most to it.



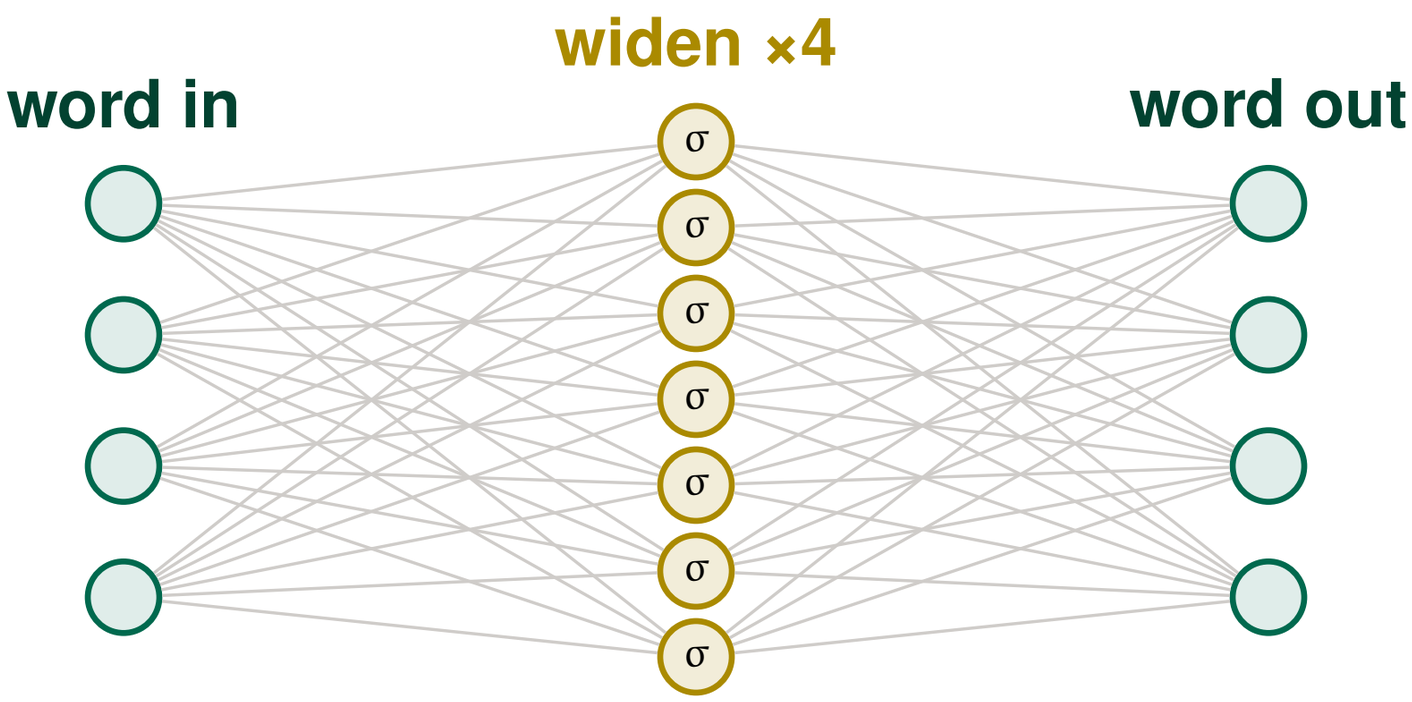
New *reads* $= 0.2\, v_{The} + 0.5\, v_{model} + 0.3\, v_{reads}$

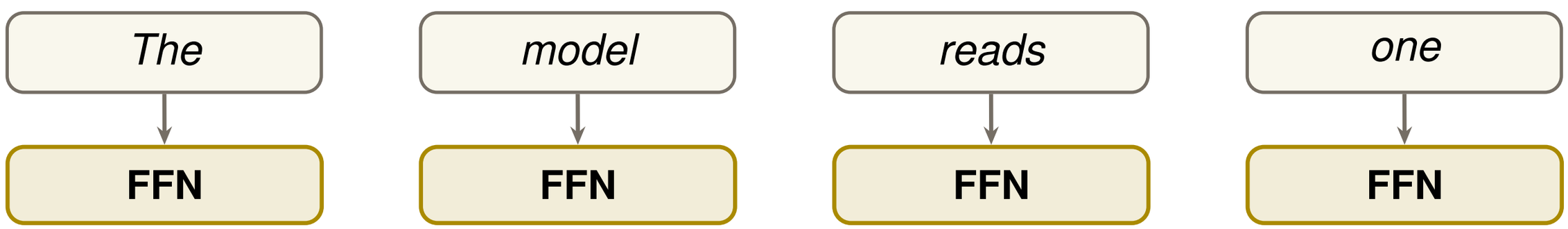
IN MAPS

$$O = \text{softmax}_{\text{row}}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V, \quad Q = UW_Q, \ldots$$



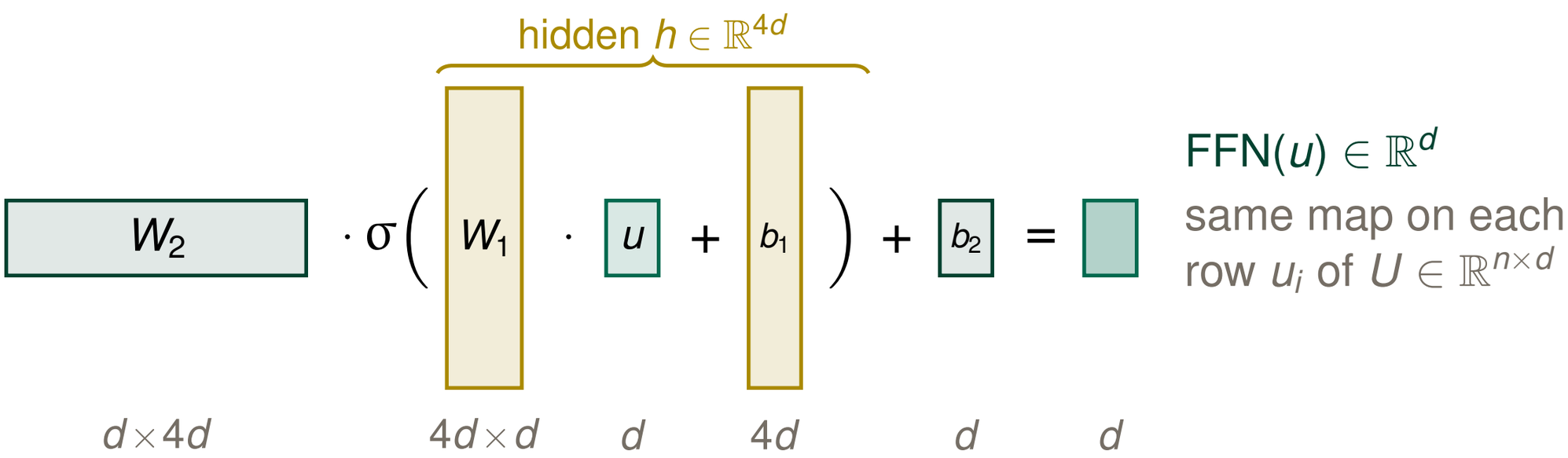
Feed-forward: each word “thinks” alone.





Same network for every word: attention mixes, FFN transforms.

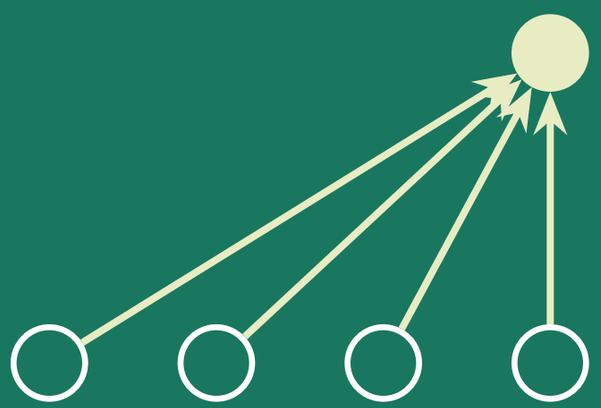
IN MAPS $\mathrm{FFN} : \mathbb{R}^d \to \mathbb{R}^d, \quad \mathrm{FFN}(u) = W_2\,\sigma(W_1 u + b_1) + b_2$



$\mathrm{FFN}(u) \in \mathbb{R}^d$
same map on each row u_i of $U \in \mathbb{R}^{n \times d}$

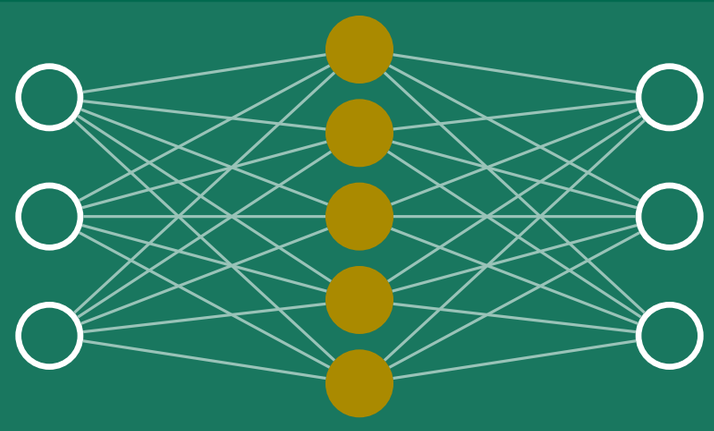
THE TAKEAWAY · 7/7

Three ideas run every LLM.



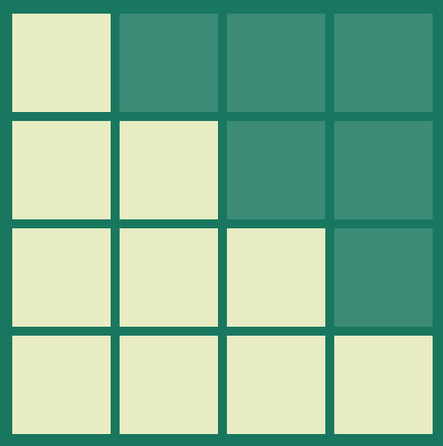
Attention

words gather from the past



Feed-forward

each word thinks alone



Masking

no peeking at the future

Full slides: link in the post.

What surprised you most? Comment below.
Follow for Part 2 and more math of AI.